A Guide to Factor-based Investing

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What Are Factors?

- Certain variables that can explain differences in the returns of stocks

- The core goal of factor investing is to understand the drivers of asset prices

Why Factors?

- To find a good portfolio in a scientific way
  - Increased profitability
  - Reduced risk

An Example: Size Factor

Firm A

Firm B

Small Cap → Higher expected return Large Cap

All else being equal (costs, profits, margins, operational leverage, etc.)

History of Factors

- CAPM (Capital Asset Pricing Model) 1960s by Markowitz, Treynor, Sharpe, Lintner, Mossin

  - Market portfolio: a portfolio of stocks weighted by the market capitalization of each stock is the most efficient portfolio

  - Efficient portfolio: one that offers the highest expected return for a given level of risk or the lowest risk for a given level of expected return

History of Factors

- Risks according to the CAPM
  - Non-systematic risk – unrewarded risk
    - Specific risk or diversifiable risk
  - Systematic risk – rewarded risk
    - The type of risk that affects all investments across the market
    - It cannot be mitigated through diversification

History of Factors

- CAPM (Capital Asset Pricing Model)

Degree of exposure to systematic market risk

$$E(R_i) = R_f + \boxed{\beta_i}\boxed{(E(R_m) - R_f)}$$

Market risk premium

$$\boxed{\alpha} = R_i - (R_f + \beta_i(E(R_m) - R_f))$$

Excess return:
Skill or luck

where:

- $E(R_i)$ is the expected return on the capital asset,
- R_f is the risk-free rate of interest (e.g., U.S. Treasury bills),
- β_i is the beta of the investment,
- $E(R_m)$ is the expected return of the market,
- $E(R_m) - R_f$ is known as the market risk premium.

An Example of CAPM

Portfolio A

50% Stocks

50% Risk free assets

Portfolio B

70% Stocks

30% Risk free assets

An Example of CAPM

Portfolio A – 7 % Return

Portfolio B – 10 % Return

3% Differences

2% is explained by beta (2/3 of performance is explained by CAPM)

1% is alpha (luck? skill? unknown factors?)

History of Factors

- Size factor (1980)
  - Small caps show better returns than their market beta
    - Market Capitalization

- Value factor (1970s and 1980s)
  - Cheap companies have higher returns than “expensive” ones
    - Price-to-Earnings Ratio (P/E)
    - Price-to-Book Ratio (P/B)

History of Factors

- 3 Factor Fama-French Model (1992)
  - Beta factor, value factor, size factor

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The Cross-Section of Expected Stock Returns

EUGENE F. FAMA, KENNETH R. FRENCH

First published: June 1992 | https://doi.org/10.1111/j.1540-6261.1992.tb04398.x |
Citations: 6,813

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ABSTRACT

Two easily measured variables, size and book-to-market equity, combine to capture the cross-sectional variation in average stock returns associated with market β, size, leverage, book-to-market equity, and earnings-price ratios. Moreover, when the tests allow for variation in β that is unrelated to size, the relation between market β and average return is flat, even when β is the only explanatory variable.

History of Factors

- 3 Factor Fama-French Model (1992)
  - It explains 90% of the differences in two diversified portfolios

- **Fama-French Three-Factor Model**: This model is a common regression model used to demonstrate the size factor. The model is:

$$R_{it} - R_{ft} = \alpha_i + \beta_{iM}(R_{Mt} - R_{ft}) + \beta_{iSMB} \cdot SMB_t + \beta_{iHML} \cdot HML_t + \epsilon_{it}$$

where:

  - $R_{it} - R_{ft}$ is the excess return of the stock or portfolio i,
  - $R_{Mt} - R_{ft}$ is the market excess return,
  - SMB_t is the size premium (small minus big),
  - HML_t is the value premium (high book-to-market minus low),

An Example of 3-factor model

Portfolio A – 7 % Return

Portfolio B – 10 % Return

3% Differences

2.7% is explained by beta (90% of performance is explained)

0.3% is alpha (luck? skill? unknown factors?)

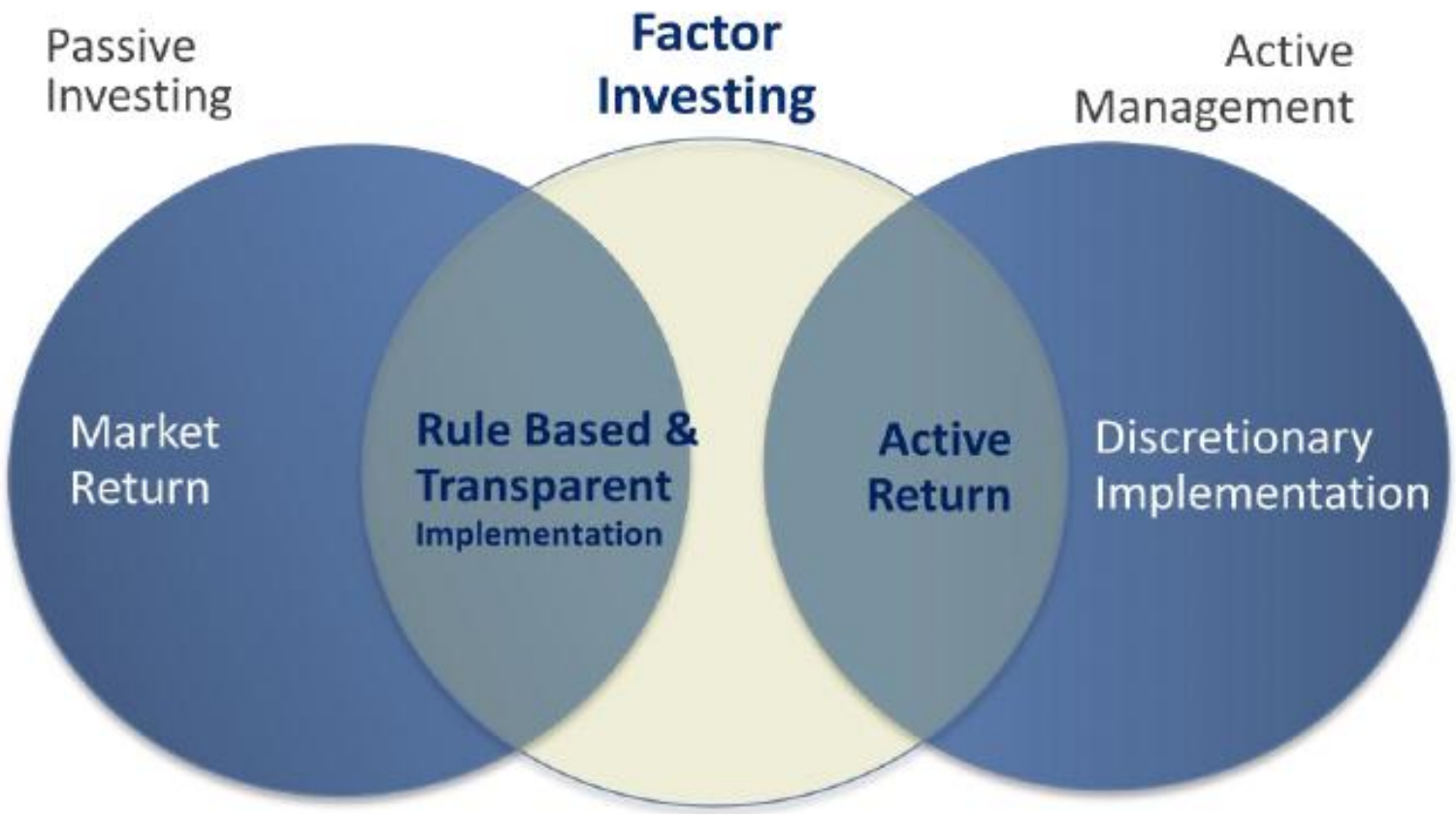
History of Factors

- Momentum factor (Jegadeesh and Titman, 1993)
  - Companies that have performed well in the recent past continue to perform better

- 4-factor model (Carhart model, 1997)
  - 3-factor + Momentum
  - Explains 95% of the differences between two diversified portfolios

- Low volatility factor (anomaly; alpha)
- Quality factor (underlying characteristics of a company)

Factors, Active and Passive Investment



Bender, J., Briand, R., Melas, D. and Subramanian, R.A., 2013. Foundations of factor investing. Available at SSRN 2543990.

Decomposing Returns to Alpha vs. Exposure to Factors

- Fund returned 116%, consistently buying small companies

- Market returned 100%

- 16% alpha
  - 15.2%: Smart beta (95% of alpha; 4-factor model criteria)
  - 0.8%: True alpha (Not explained by factors)

Characteristics of Factors

- Persistent (across different time periods)
- Pervasive (across different countries)
- Robust (across variations of formulation; e.g. P/B ratio, EBITA)
- Explainable (compensation to risk vs. market inefficiency)
- Investable

Implement Factor Exposures to Portfolio

- Long only
  - Sorted portfolio
- Long/Short position based on a quantile portfolio
- Factor-tilts

Stock ID	P/B Ratio
S2	0.5
S5	0.8
S3	1.1
S1	1.8
S4	2.4

- Smart beta (ETFs)
  - Smart-beta ETFs allow investors to directly invest in particular factors (value, low volatility, etc.)
    - E.g. iShares Edge MSCI USA Momentum Factor ETF (MTUM)

How to Find Good Factors

The Workflow of Factor Finding

- Step1. Universe filtering
- Step2. Sorting based on a particular factor (e.g., size, book-to-market ratio)
- Step3. Quantile portfolio construction (J=2, J=3, J=5 or J=10 portfolios)
- Step4. Weighting schemes (Equally weighted, value weighted, risk parity)
- Step5. Report the returns of the portfolios (T-test)

Step1. Universe Filtering

- (Terminology) Universe
  - Equities composing the benchmark (e.g. S&P500, KOSPI 200)

- Why do we need universe filters?
  - Reduce potential “Luck” components
  - Building a tradable stock universe (Backtesting = Real Investments)
    - Transaction Costs
    - High Turnovers
    - Timing Loss

Step1. Universe Filtering

- Examples of universe filters
  - Market capitalization
  - Trading dollar volume
  - Days from IPO

Step2. Sorting Based on a Particular Factor Criteria

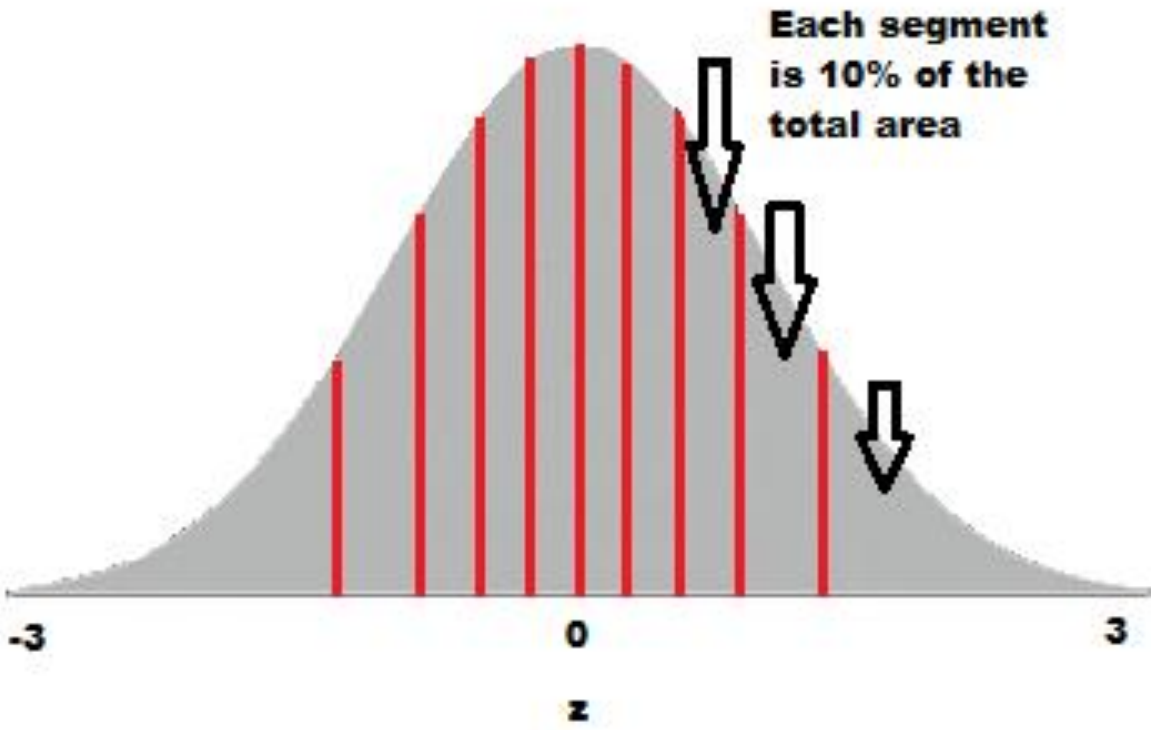
- Rank firms according to a particular criterion at formation period
  - e.g., momentum, size, book-to-market ratio
  - momentum: sort portfolio based on 12-1M return

https://quant.stackexchange.com/questions/26382/how-to-calculate-monthly-return-from-a-momentum-strategy-with-overlapping-holdin

Step3. Quantile Portfolio Construction

- Form J ≥ 2 portfolios (i.e., homogeneous groups) consisting of the same number of stocks according to the ranking

  - Ex. Decile



Step4. Weighting Schemes

- Equally weighted

$$W_i = \frac{1}{N}$$

$$N = Total\ number\ of\ securities\ in\ the\ quantile$$
$$W_i = weight\ of\ security$$

  - May have higher transaction costs due to frequent rebalancing required to maintain equal weights
  - Have more weights to small-cap than value weighted

Step4. Weighting Schemes

- Value weighted

$$W_i = \frac{Market\ Cap\ of\ Security\ i}{Total\ Market\ Cap\ of\ the\ Quantile}$$

  - Accounts for market capitalization and weights stocks based on their market value, which means larger companies have a larger impact on the portfolio
  - Can lead to concentration risk, as the portfolio may be heavily weighted towards a few large companies or sectors

Step4. Weighting Schemes

- Risk parity

$$W_i = \frac{\sigma_i^{-1}}{\Sigma \sigma_j^{-1}}$$

Assuming that the assets are uncorrelated

$$\sigma_i = Standard\ deviation\ of\ security\ i$$

  - Designed to provide equal risk exposure across asset classes, which can lead to a more stable portfolio in volatile market conditions
  - Can potentially provide better risk-adjusted returns compared to other weighting schemes
  - Diversifies the portfolio beyond traditional asset classes and can include alternative investments such as commodities or real estate

Step5. Report the Returns of the Portfolios

- The core goal of factors is to explain the cross-section of stock returns

- An anomaly is identified if the t-test between the first ($j = 1$) and the last group ($j = J$) shows a significant difference in average returns

Step5. Report the Returns of the Portfolios

Table IV

Momentum Strategy Returns

This table reports the mean quintile portfolio returns based on the past one-week, two-week, three-week, four-week, and one-to-four-week return measures. The mean returns are the time-series averages of weekly value-weighted portfolio excess returns. *, **, and *** denote significance at the 10%, 5%, and 1% levels.

	Quintiles					
	1	2	3	4	5	5–1
r 1,0	**Low**				**High**	
Mean	−0.002	0.000	0.010	0.036**	0.023**	0.025**
t(Mean)	(−0.19)	(0.04)	(1.45)	(2.52)	(2.03)	(2.19)
r 2,0	**Low**				**High**	
Mean	0.000	0.005	0.009	0.017**	0.031***	0.031***
t(Mean)	(0.01)	(0.66)	(1.33)	(2.15)	(2.93)	(2.90)
r 3,0	**Low**				**High**	
Mean	0.005	0.002	0.016*	0.017**	0.036***	0.031***
t(Mean)	(0.60)	(0.28)	(1.94)	(2.30)	(3.21)	(2.65)
r 4,0	**Low**				**High**	
Mean	0.002	0.005	0.009	0.020**	0.025**	0.022**
t(Mean)	(0.30)	(0.66)	(1.28)	(2.45)	(2.32)	(2.26)
r 4,1	**Low**				**High**	
Mean	0.003	0.007	0.021**	0.011	0.020**	0.017*
t(Mean)	(0.35)	(0.94)	(2.34)	(1.51)	(2.02)	(1.82)

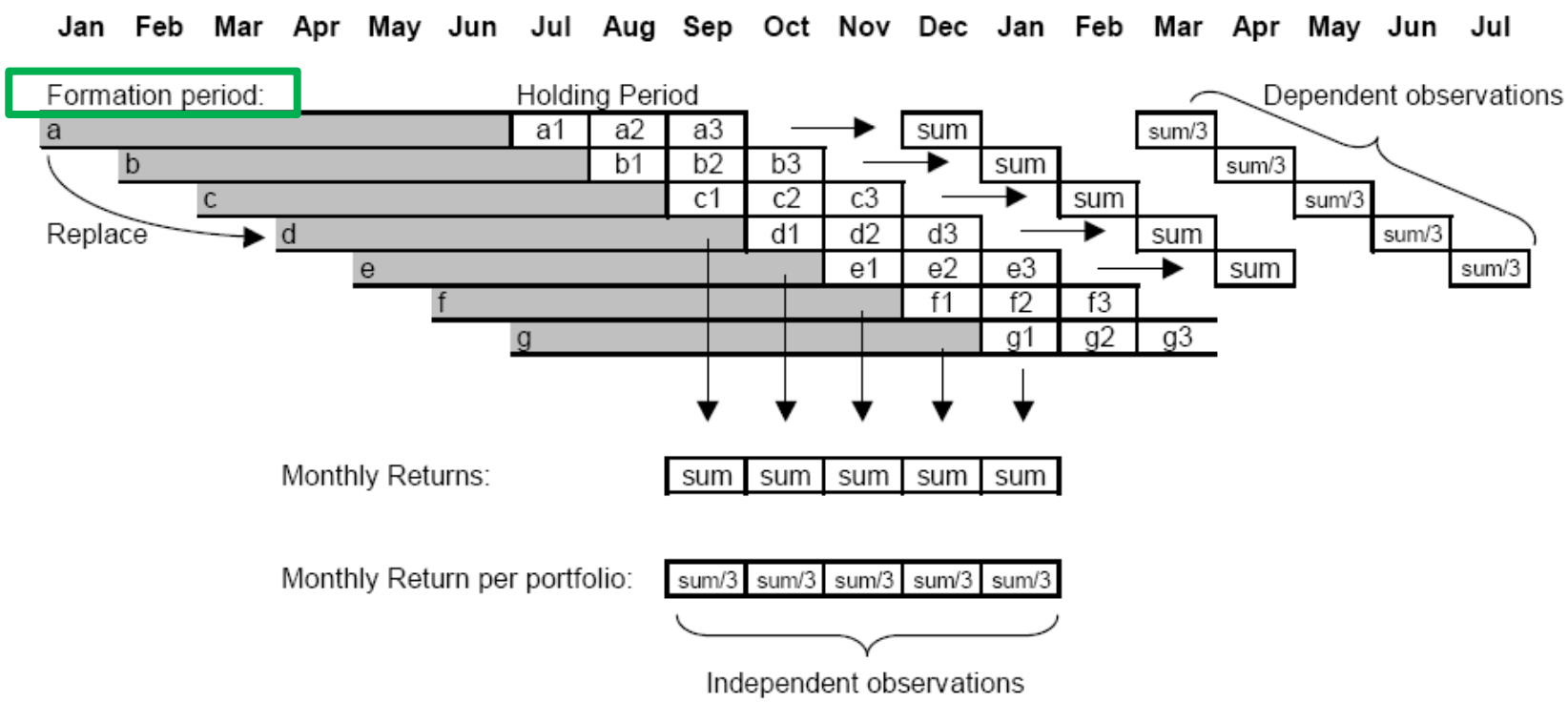
Step5. Report the Returns of the Portfolios

- Evaluate long only position
  - In most cases, portfolios take long only positions, so we cannot exploit (long - short factor) returns
    - Long only position = market factor + factor tilts

  - Benchmark for long only position
    - Market Cap Weighted
    - Equally Weighted
    - Risk Parity

Reducing Rebalancing Luck

- Formation periods: 1Y
- Holding periods: 1Y
- Rebalancing periods: 1M



The Workflow of Factor Models

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- Step2. Sorting based on a particular factor (e.g., size, book-to-market ratio)
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- Step5. Report the returns of the portfolios (T-test)

Underlying Mechanisms of Risk Premiums

- **Type1. Risk Compensation**
  - Compensation from exposures to systemic risk

- **Type2. Mispricing**
  - Efficient Market Hypothesis (EMH)
    - Null Hypothesis: News is rapidly and fully incorporated in prices

Examples of Common Factor Groups

Systematic Factors	What It is	Commonly Captured by
Value	➢ Captures excess returns to stocks that have low prices relative to their fundamental value	➢ Book to price, earnings to price, book value, sales, earnings, cash earnings, net profit, dividends, cash flow
Low Size (Small Cap)	➢ Captures excess returns of smaller firms (by market capitalization) relative to their larger counterparts	➢ Market capitalization (full or free float)
Momentum	➢ Reflects excess returns to stocks with stronger past performance	➢ Relative returns (3-mth, 6-mth, 12-mth, sometimes with last 1 mth excluded), historical alpha
Low Volatility	➢ Captures excess returns to stocks with lower than average volatility, beta, and/or idiosyncratic risk	➢ Standard deviation (1-yr, 2-yrs, 3-yrs), Downside standard deviation, standard deviation of idiosyncratic returns, Beta
Dividend Yield	➢ Captures excess returns to stocks that have higher-than-average dividend yields	➢ Dividend yield
Quality	➢ Captures excess returns to stocks that are characterized by low debt, stable earnings growth, and other "quality" metrics	➢ ROE, earnings stability, dividend growth stability, strength of balance sheet, financial leverage, accounting policies, strength of management, accruals, cash flows

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Underlying Mechanisms of Factors

Factor Group	Risk Compensation	Mispricing
High Value	Higher financial distress risks and sensitivity to economic cycles.	Overreactions to negative news or earnings disappointments.
Small Size	Higher operational and financial risks due to smaller scale.	Overlooked by analysts and investors, leading to underrepresentation in the market.
High Momentum	n.a.	Slow reaction to new information and trend persistence due to behavioral biases.
High Yield	Risk of dividend cuts if high yields are due to declining prices.	High yields might distract from underlying company issues, leading to mispricing.
Low Volatility	Underperformance in bull markets (missing out on bigger gains).	Overconfidence in volatile stocks might lead to undervaluation of stable, low-volatility stocks.
High Quality	Premium pricing may lead to overvaluation in market downturns.	Underestimation of the benefits of stability and consistent performance in quality stocks.

Characteristic Models

What Are Characteristics?

- **Factor Models**
  - Factor models explain the returns of a broad set of securities through its exposure to various risk factors

- **Characteristics Models**
  - Characteristic models focus on the individual attributes or characteristics of the securities themselves rather than on broader risk factors

Why Characteristic Models? Limitations of Sorted Portfolios

- The sorting criterion could have a non-monotonic (non-linear) impact on returns

- Sorted portfolios may capture not only the priced risk associated with the characteristic, but also some unpriced risk
  - Need to control other factors to isolate the effect of a single factor

Factor Model: Cross-sectional regression

$X_{t,n,k}$

Exposure of factor k to stock n at time t

Factor Model: Cross-sectional regression

Return of Factor k at time t (Common variables across different n)

Return $\boldsymbol{r_{t,n}} = \Sigma \, \mathrm{x}_{\,t,n,k} f_{t,k} + \varepsilon_{t,n}$

Exposure of factor k to stock n at time t

Factor Model: Cross-sectional regression

Return of Factor k at time t (Common variables across different n)

Return $r_{t,n} = \Sigma\, x_{t,n,k} f_{t,k} + \varepsilon_{t,n}$

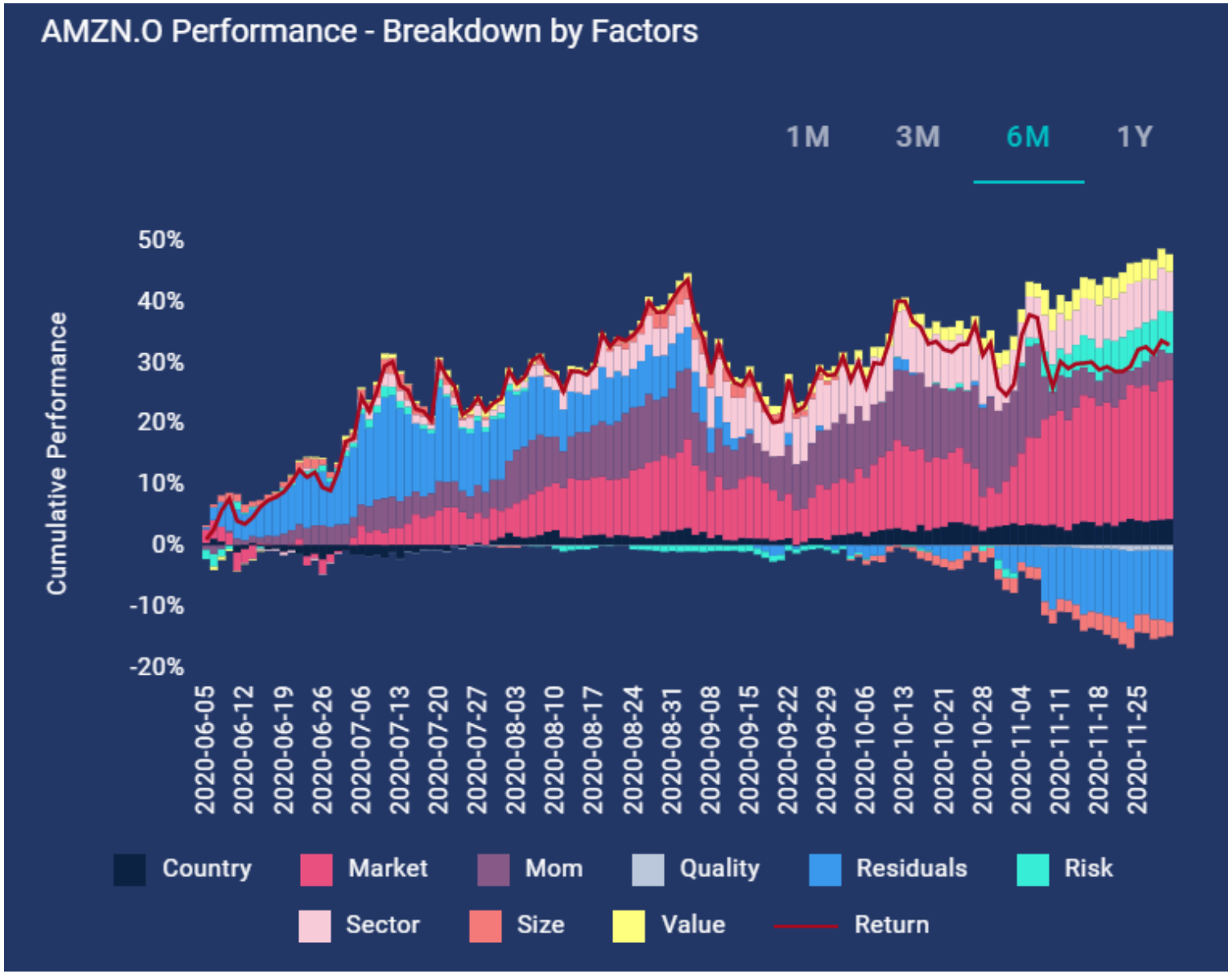
Exposure of factor k to stock n at time t

Factor returns to infer

Residuals to minimize

$$
\begin{aligned}
R_{s_1} &= X_{s_1 f_1} F_{f_1} + X_{s_1 f_2} F_{f_2} + \cdots + X_{s_1 f_{100}} F_{f_{100}} + \varepsilon_{s_1} \\
R_{s_2} &= X_{s_2 f_1} F_{f_1} + X_{s_2 f_2} F_{f_2} + \cdots + X_{s_2 f_{100}} F_{f_{100}} + \varepsilon_{s_2} \\
&\cdots \\
R_{s_{2999}} &= X_{s_{2999} f_1} F_{f_1} + X_{s_{2999} f_2} F_{f_2} + \cdots + X_{s_{2999} f_{100}} F_{f_{100}} + \varepsilon_{s_{2999}} \\
R_{s_{3000}} &= X_{s_{3000} f_1} F_{f_1} + X_{s_{3000} f_2} F_{f_2} + \cdots + X_{s_{3000} f_{100}} F_{f_{100}} + \varepsilon_{s_{3000}}
\end{aligned}
$$

Factor Model: Cross-sectional regression



Connections between Asset Pricing and Factor Models

Firm characteristics (e.g. market capitalization, accounting ratios)

$X_{t,n}$

Connections between Asset Pricing and Factor Models

Firm characteristics (e.g. market capitalization, accounting ratios)

Future return $\boldsymbol{r}_{t+1,n} = f(\mathrm{x}_{t,n}) + \varepsilon_{t+1,n}$

Model
(e,g, linear model, neural network)

Connections between Asset Pricing and Factor Models

- **Factor Model: Cross-sectional regression (Explanation)**

Return of Factor k at time t (Common variables across different n)

Return $\boldsymbol{r_{t,n}} = \Sigma\, x_{t,n,k} f_{t,k} + \varepsilon_{t,n}$

Exposure of factor k to stock n

- **Asset Pricing Model (Prediction)**

Firm characteristics: Exposure to macro-economic factors (factor loadings)

Future return $\boldsymbol{r_{t+1,n}} = f(\, x_{t,n}\,) + \varepsilon_{t+1,n}$

Model

(e,g, linear model, neural network)

Empirical Asset Pricing via Machine Learning

$x_{t,n}$ Firm characteristics: Exposure to macro-economic factors (factor loadings)

Table A.6: Details of the Characteristics

No.	Acronym	Firm characteristic	Paper's author(s)	Year, Journal	Data Source	Frequency
1	absacc	Absolute accruals	Bandyopadhyay, Huang & Wirjanto	2010, WP	Compustat	Annual
2	acc	Working capital accruals	Sloan	1996, TAR	Compustat	Annual
3	aeavol	Abnormal earnings announcement volume	Lerman, Livnat & Mendenhall	2007, WP	Compustat+CRSP	Quarterly
4	age	# years since first Compustat coverage	Jiang, Lee & Zhang	2005, RAS	Compustat	Annual
5	agr	Asset growth	Cooper, Gulen & Schill	2008, JF	Compustat	Annual
6	baspread	Bid-ask spread	Amihud & Mendelson	1989, JF	CRSP	Monthly
7	beta	Beta	Fama & MacBeth	1973, JPE	CRSP	Monthly
8	betasq	Beta squared	Fama & MacBeth	1973, JPE	CRSP	Monthly
9	bm	Book-to-market	Rosenberg, Reid & Lanstein	1985, JPM	Compustat+CRSP	Annual
10	bm_ia	Industry-adjusted book to market	Asness, Porter & Stevens	2000, WP	Compustat+CRSP	Annual
11	cash	Cash holdings	Palazzo	2012, JFE	Compustat	Quarterly
12	cashdebt	Cash flow to debt	Ou & Penman	1989, JAE	Compustat	Annual
13	cashpr	Cash productivity	Chandrashekar & Rao	2009, WP	Compustat	Annual
14	cfp	Cash flow to price ratio	Desai, Rajgopal & Venkatachalam	2004, TAR	Compustat	Annual
15	cfp_ia	Industry-adjusted cash flow to price ratio	Asness, Porter & Stevens	2000, WP	Compustat	Annual
16	chatoia	Industry-adjusted change in asset turnover	Soliman	2008, TAR	Compustat	Annual
17	chcsho	Change in shares outstanding	Pontiff & Woodgate	2008, JF	Compustat	Annual
18	chempia	Industry-adjusted change in employees	Asness, Porter & Stevens	1994, WP	Compustat	Annual
19	chinv	Change in inventory	Thomas & Zhang	2002, RAS	Compustat	Annual
20	chmom	Change in 6-month momentum	Gettleman & Marks	2006, WP	CRSP	Monthly
21	chpmia	Industry-adjusted change in profit margin	Soliman	2008, TAR	Compustat	Annual
22	chtx	Change in tax expense	Thomas & Zhang	2011, JAR	Compustat	Quarterly
23	cinvest	Corporate investment	Titman, Wei & Xie	2004, JFQA	Compustat	Quarterly
24	convind	Convertible debt indicator	Valta	2016, JFQA	Compustat	Annual
25	currat	Current ratio	Ou & Penman	1989, JAE	Compustat	Annual
26	depr	Depreciation / PP&E	Holthausen & Larcker	1992, JAE	Compustat	Annual
27	divi	Dividend initiation	Michaely, Thaler & Womack	1995, JF	Compustat	Annual
28	divo	Dividend omission	Michaely, Thaler & Womack	1995, JF	Compustat	Annual
29	dolvol	Dollar trading volume	Chordia, Subrahmanyam & Anshuman	2001, JFE	CRSP	Monthly
30	dy	Dividend to price	Litzenberger & Ramaswamy	1982, JF	Compustat	Annual
31	ear	Earnings announcement return	Kishore, Brandt, Santa-Clara & Venkatachalam	2008, WP	Compustat+CRSP	Quarterly

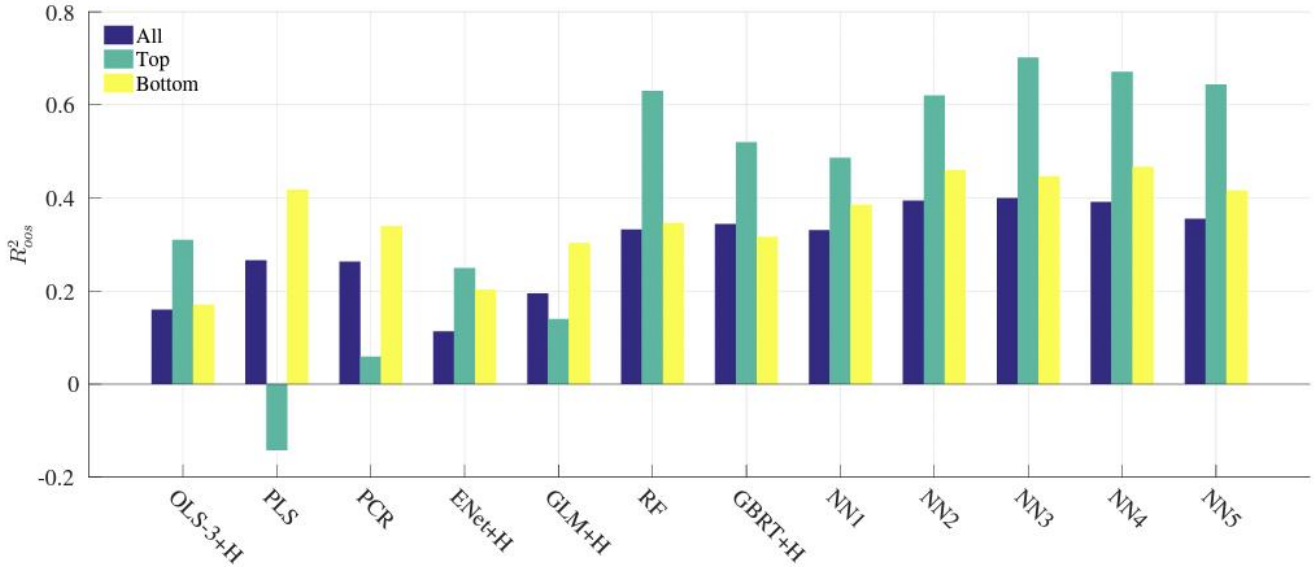
Gu, S., Kelly, B. and Xiu, D., 2020. Empirical asset pricing via machine learning. *The Review of Financial Studies*, *33*(5), pp.2223-2273.

Empirical Asset Pricing via Machine Learning

$f(x_{t,n})$: $Model$ $(e, g, linear\ model, neural\ network)$

Table 1: Monthly Out-of-sample Stock-level Prediction Performance (Percentage R^2_{oos})

	OLS +H	OLS-3 +H	PLS	PCR	ENet +H	GLM +H	RF	GBRT +H	NN1	NN2	NN3	NN4	NN5
All	-3.46	0.16	0.27	0.26	0.11	0.19	0.33	0.34	0.33	0.39	0.40	0.39	0.36
Top 1000	-11.28	0.31	-0.14	0.06	0.25	0.14	0.63	0.52	0.49	0.62	0.70	0.67	0.64
Bottom 1000	-1.30	0.17	0.42	0.34	0.20	0.30	0.35	0.32	0.38	0.46	0.45	0.47	0.42



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Cross-sectional Machine Learning Portfolios

Table 7: Performance of Machine Learning Portfolios

	OLS-3+H				PLS				PCR			
	Pred	Avg	Std	SR	Pred	Avg	Std	SR	Pred	Avg	Std	SR
Low(L)	-0.17	0.40	5.90	0.24	-0.83	0.29	5.31	0.19	-0.68	0.03	5.98	0.02
2	0.17	0.58	4.65	0.43	-0.21	0.55	4.96	0.38	-0.11	0.42	5.25	0.28
3	0.35	0.60	4.43	0.47	0.12	0.64	4.63	0.48	0.19	0.53	4.94	0.37
4	0.49	0.71	4.32	0.57	0.38	0.78	4.30	0.63	0.42	0.68	4.64	0.51
5	0.62	0.79	4.57	0.60	0.61	0.77	4.53	0.59	0.62	0.81	4.66	0.60
6	0.75	0.92	5.03	0.63	0.84	0.88	4.78	0.64	0.81	0.81	4.58	0.61
7	0.88	0.85	5.18	0.57	1.06	0.92	4.89	0.65	1.01	0.87	4.72	0.64
8	1.02	0.86	5.29	0.56	1.32	0.92	5.14	0.62	1.23	1.01	4.77	0.73
9	1.21	1.18	5.47	0.75	1.66	1.15	5.24	0.76	1.52	1.20	4.88	0.86
High(H)	1.51	1.34	5.88	0.79	2.25	1.30	5.85	0.77	2.02	1.25	5.60	0.77
H-L	1.67	0.94	5.33	0.61	3.09	1.02	4.88	0.72	2.70	1.22	4.82	0.88

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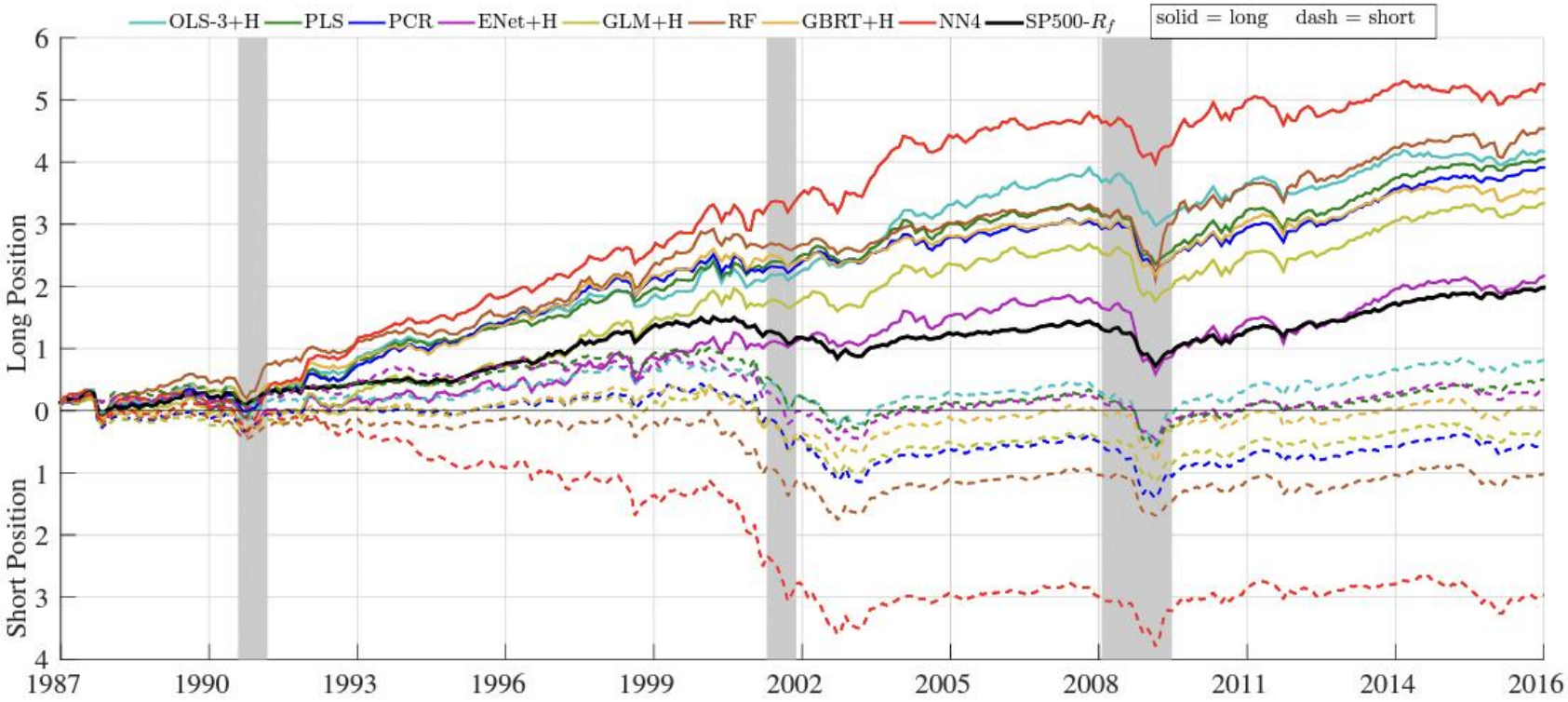
Machine Learning Portfolios

Figure 9: Cumulative Return of Machine Learning Portfolios



Note: Cumulative log returns of portfolios sorted on out-of-sample machine learning return forecasts. The solid and dash lines represent long (top decile) and short (bottom decile) positions, respectively. The shaded periods show NBER recession dates. All portfolios are value weighted.

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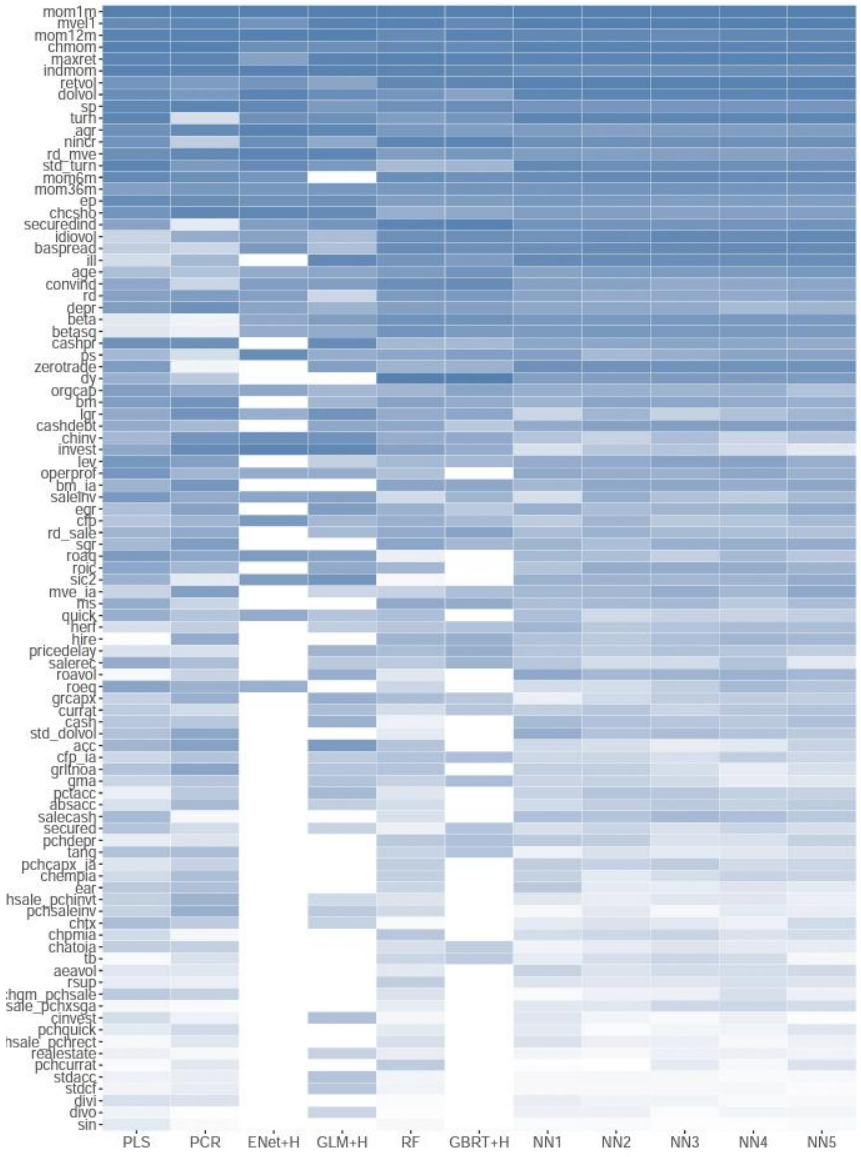
Which Characteristics Matter

Figure 5: Characteristic Importance

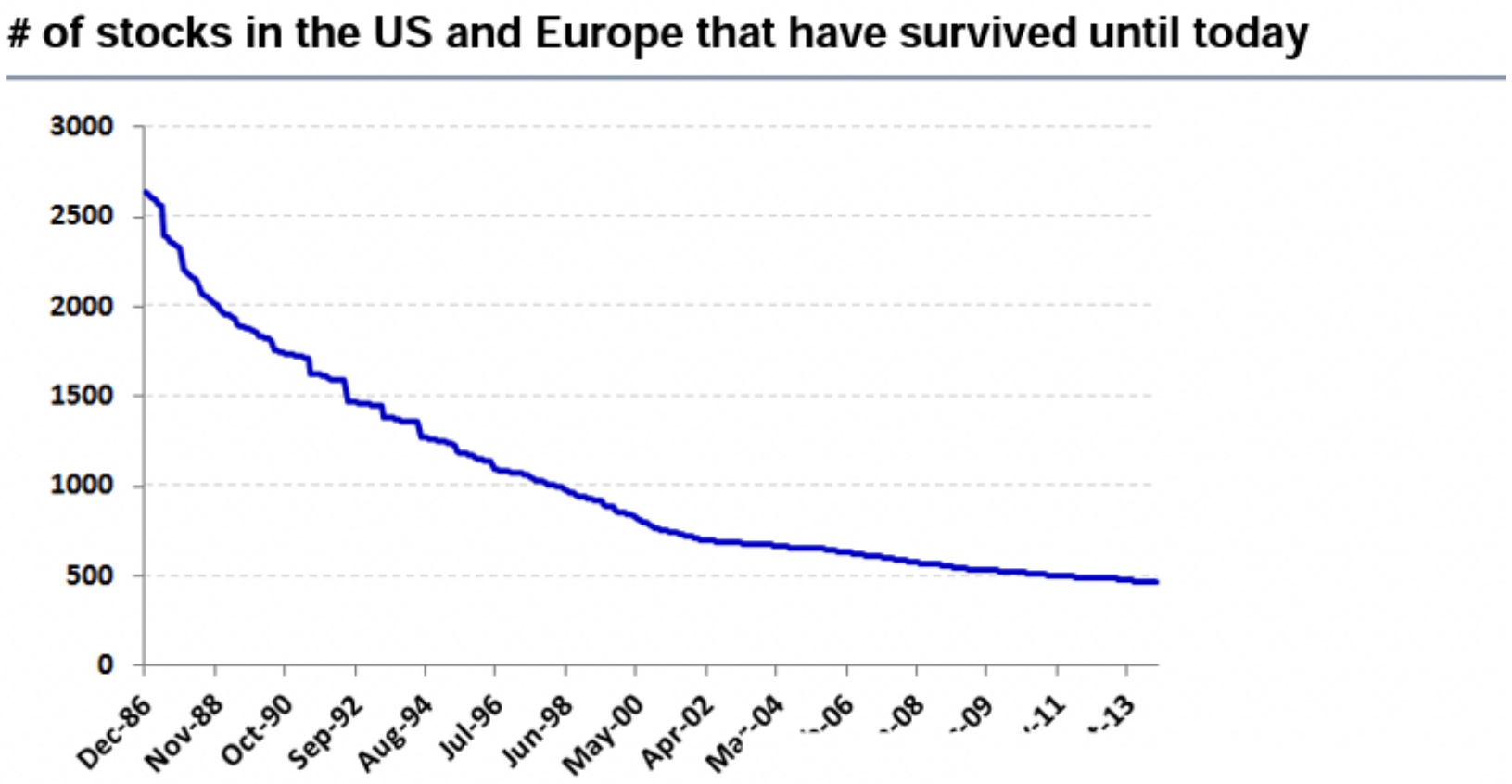


- Not all factor exposures are expected to earn a return premium over the long term
- Factor momentum, factor timing

Gu, S., Kelly, B. and Xiu, D., 2020. Empirical asset pricing via machine learning. *The Re view of Financial Studies*, *33*(5), pp.2223-2273.

Robustness Checks

Robustness Check1. Survivorship bias



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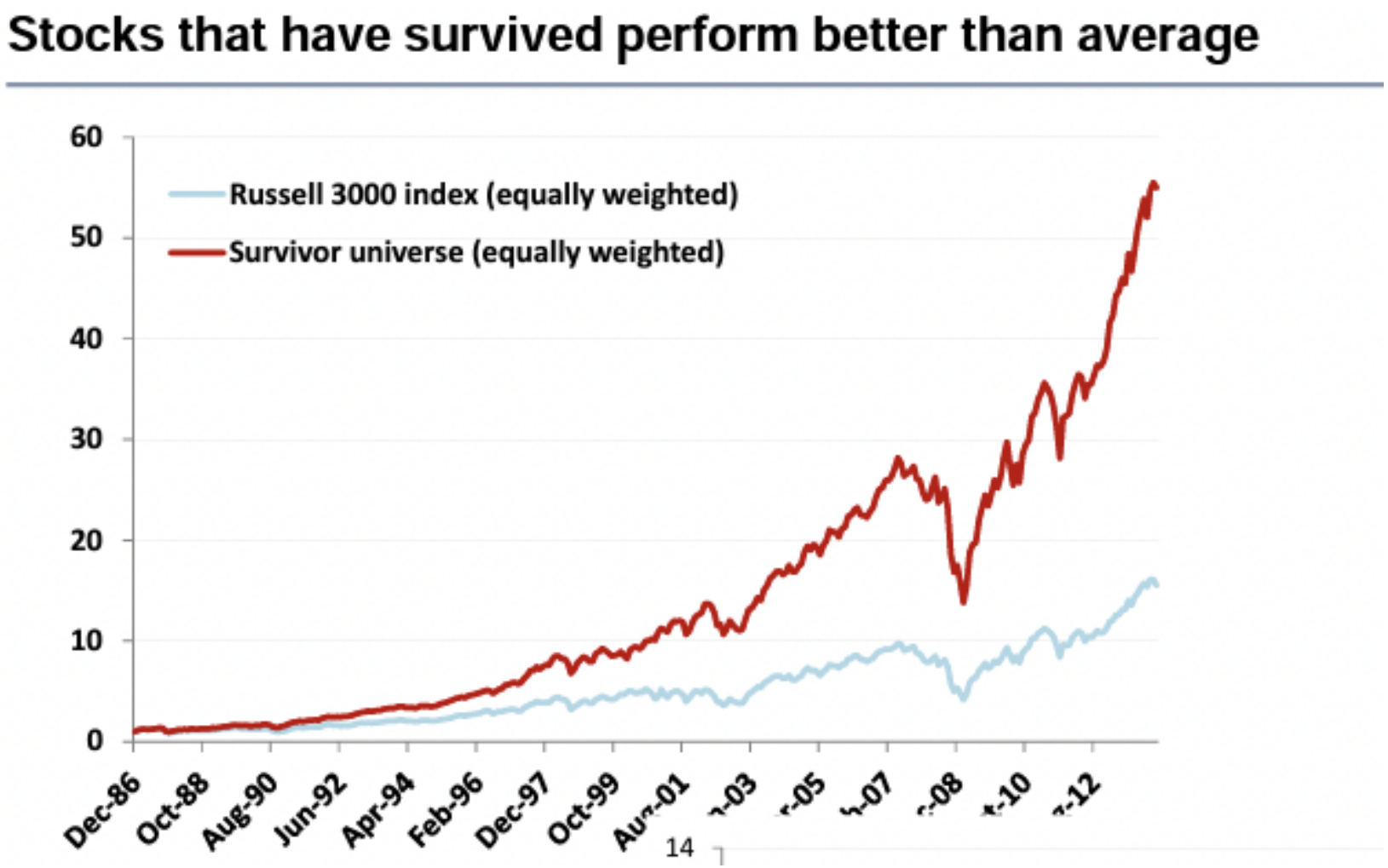
Robustness Check1. Survivorship bias



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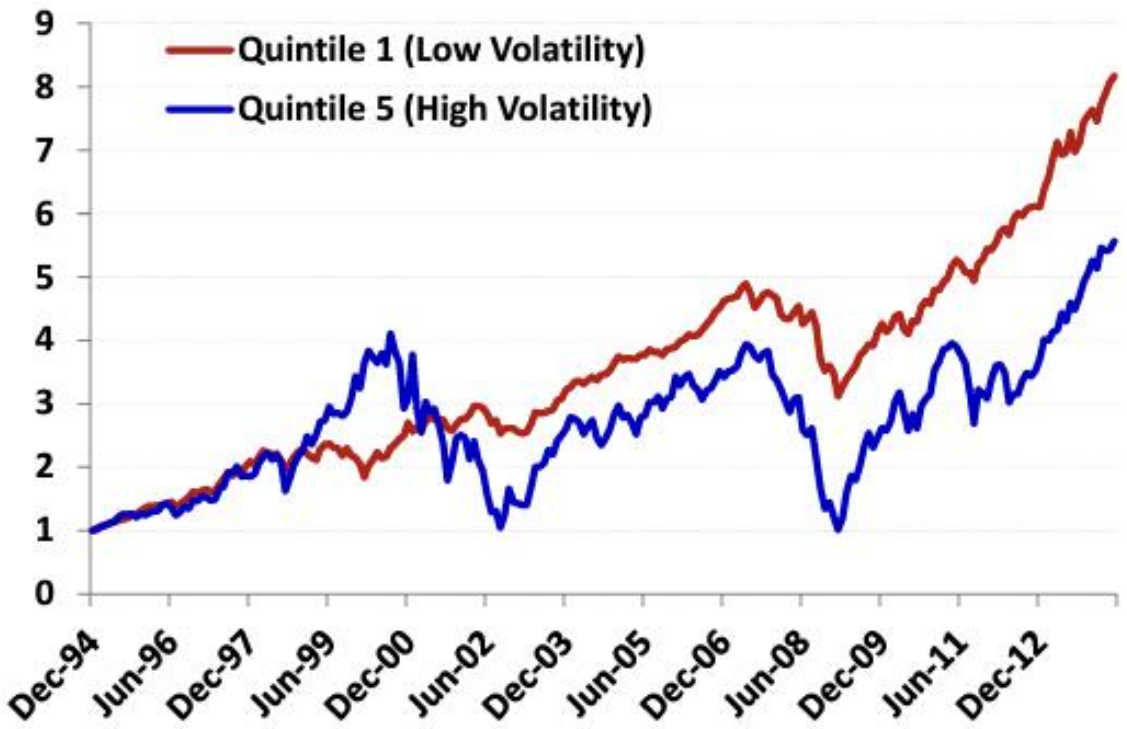
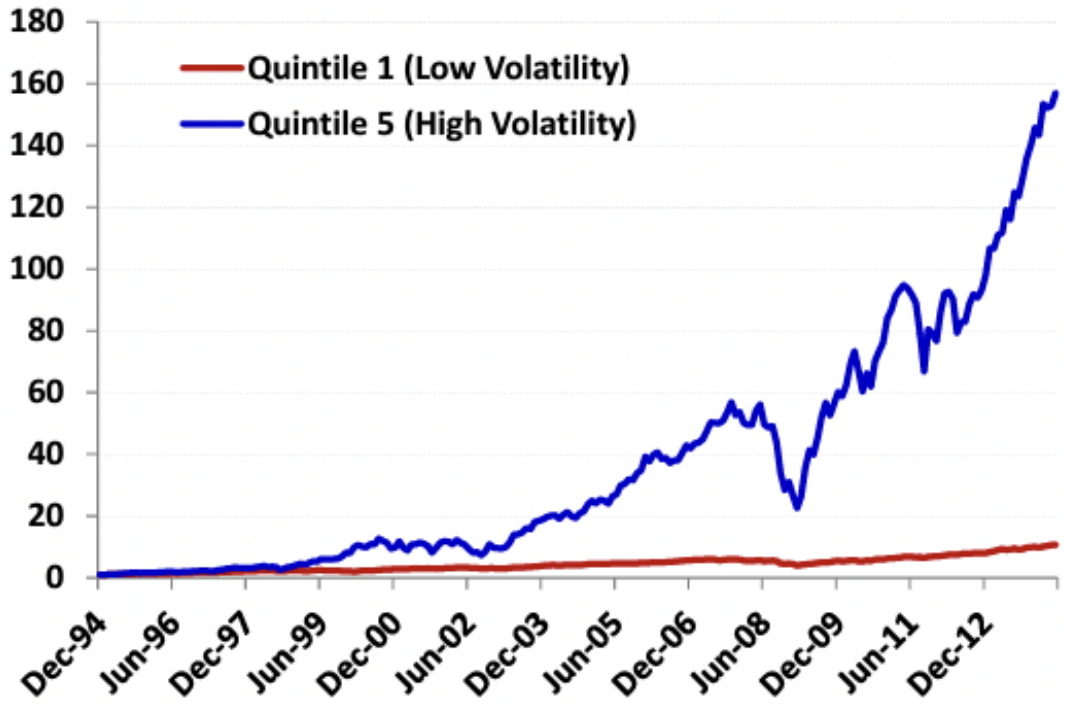
Robustness Check1. Survivorship bias

Low volatility factor on the proper S&P 500 universe



Low volatility factor performance on the current S&P 500 index constituents



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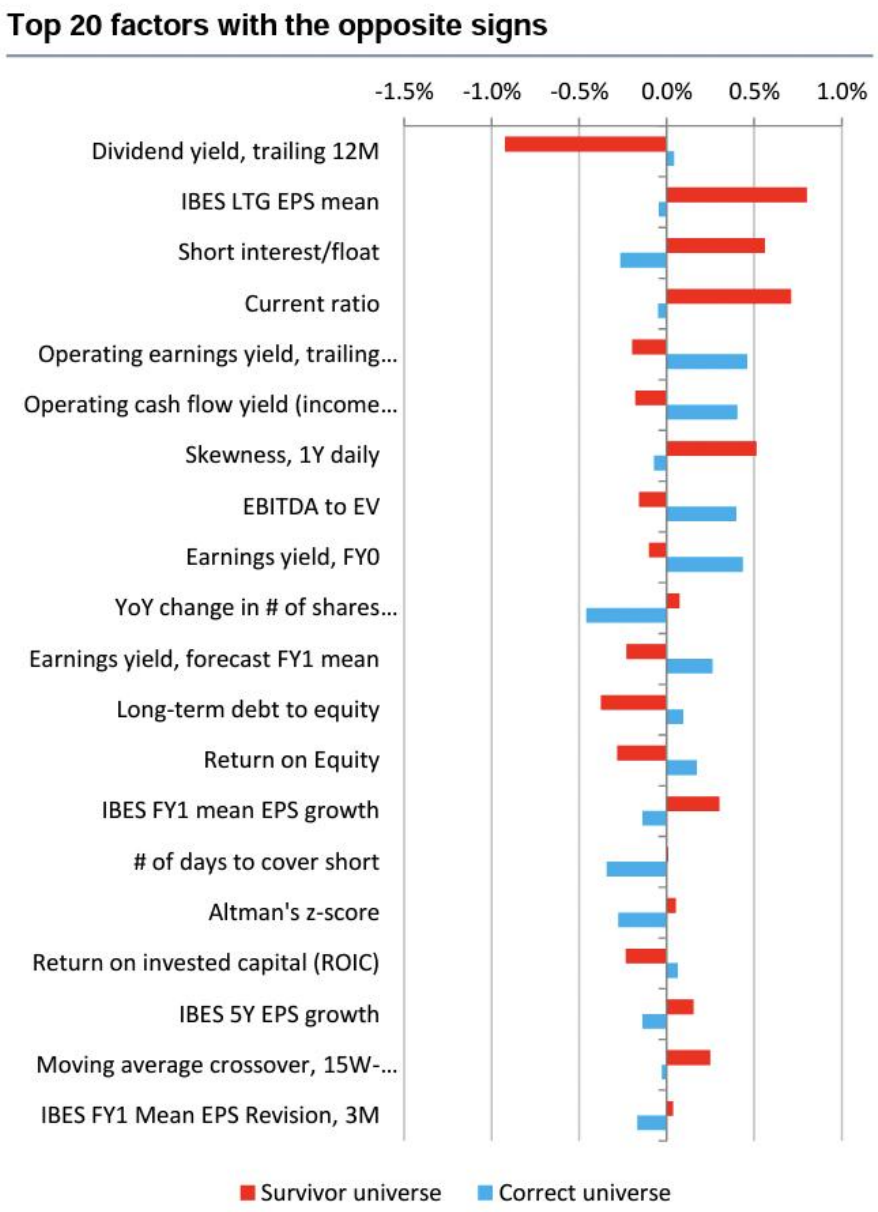

- 1/3 of factors have the opposite signs

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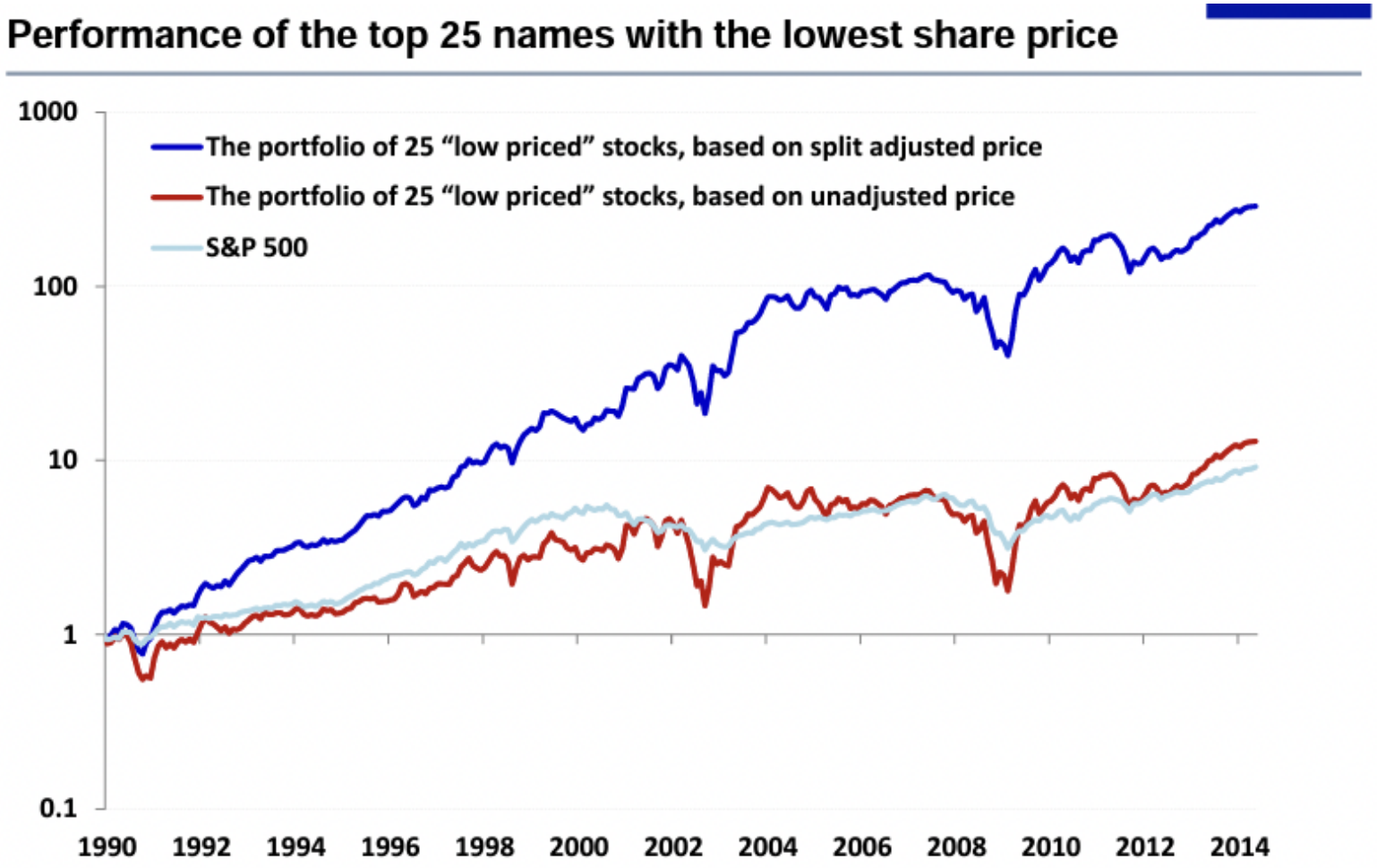
Robustness Check2. Look-ahead bias

- Using data that were unknown



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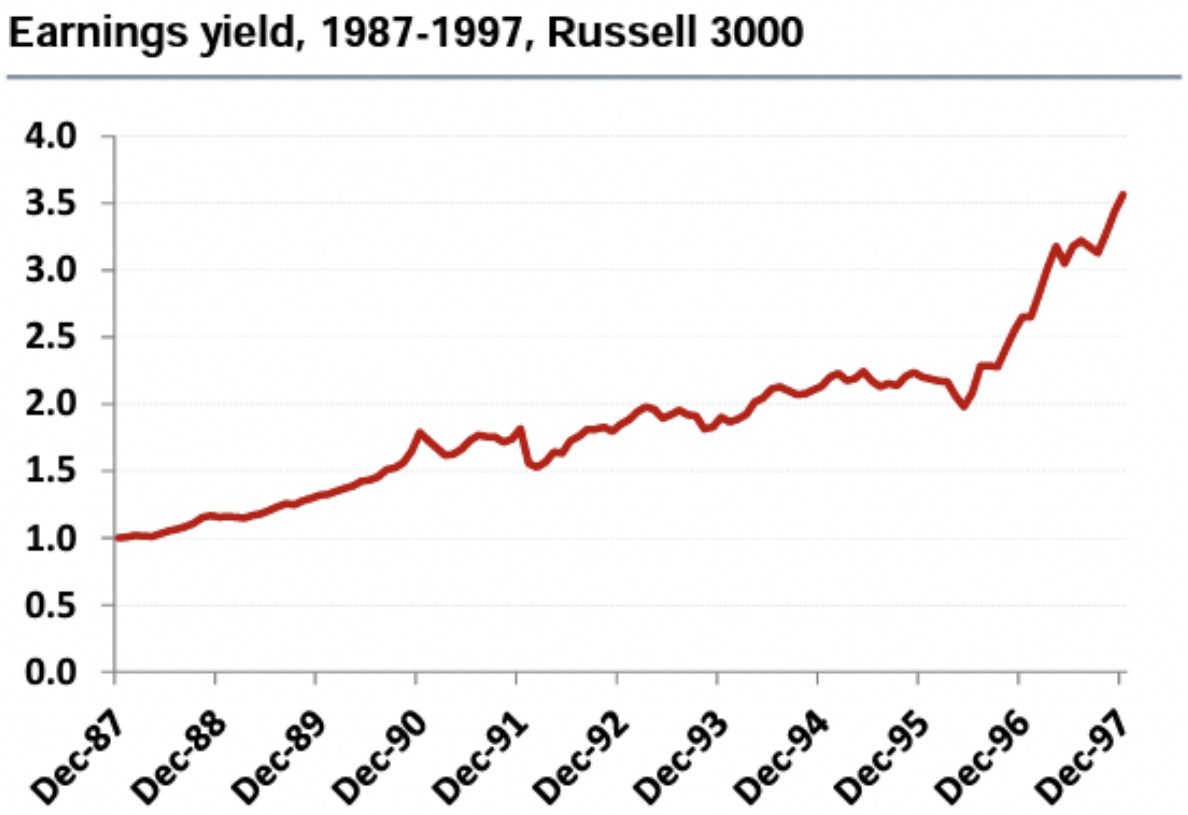
Robustness Check3. How long is long enough?



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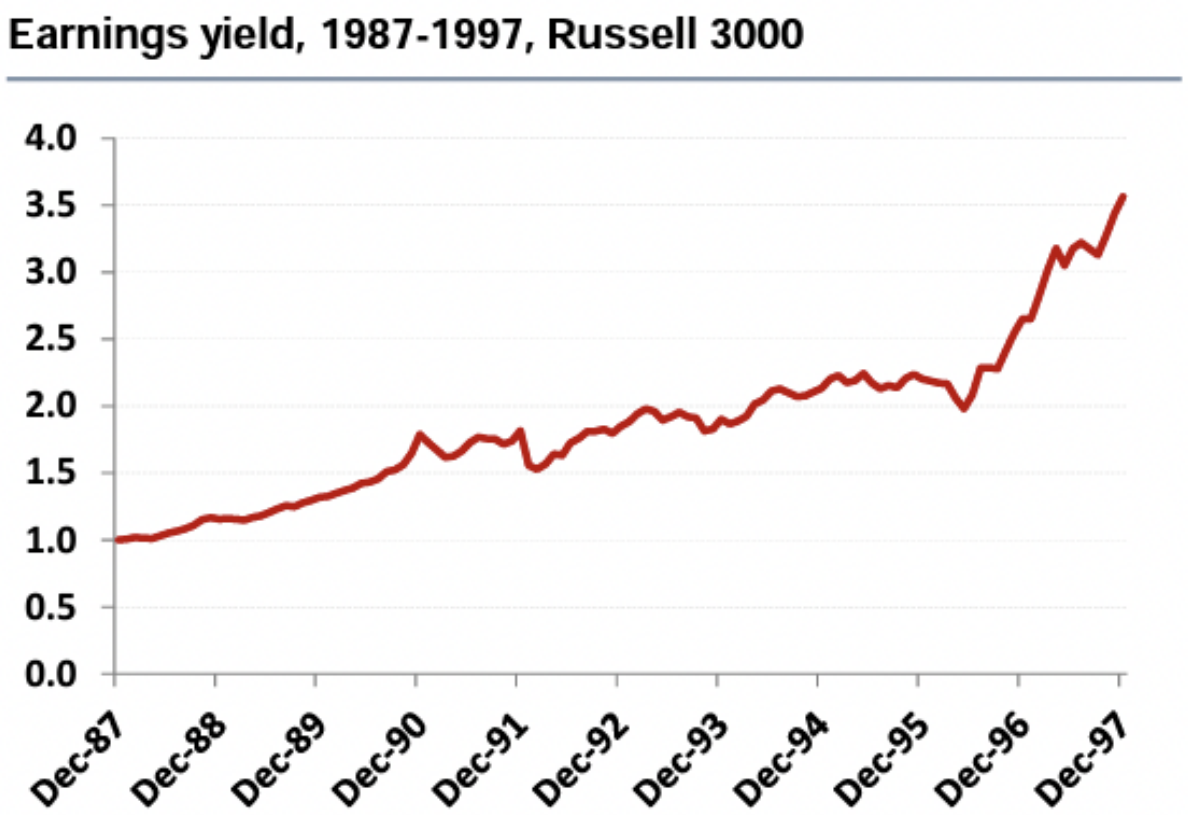

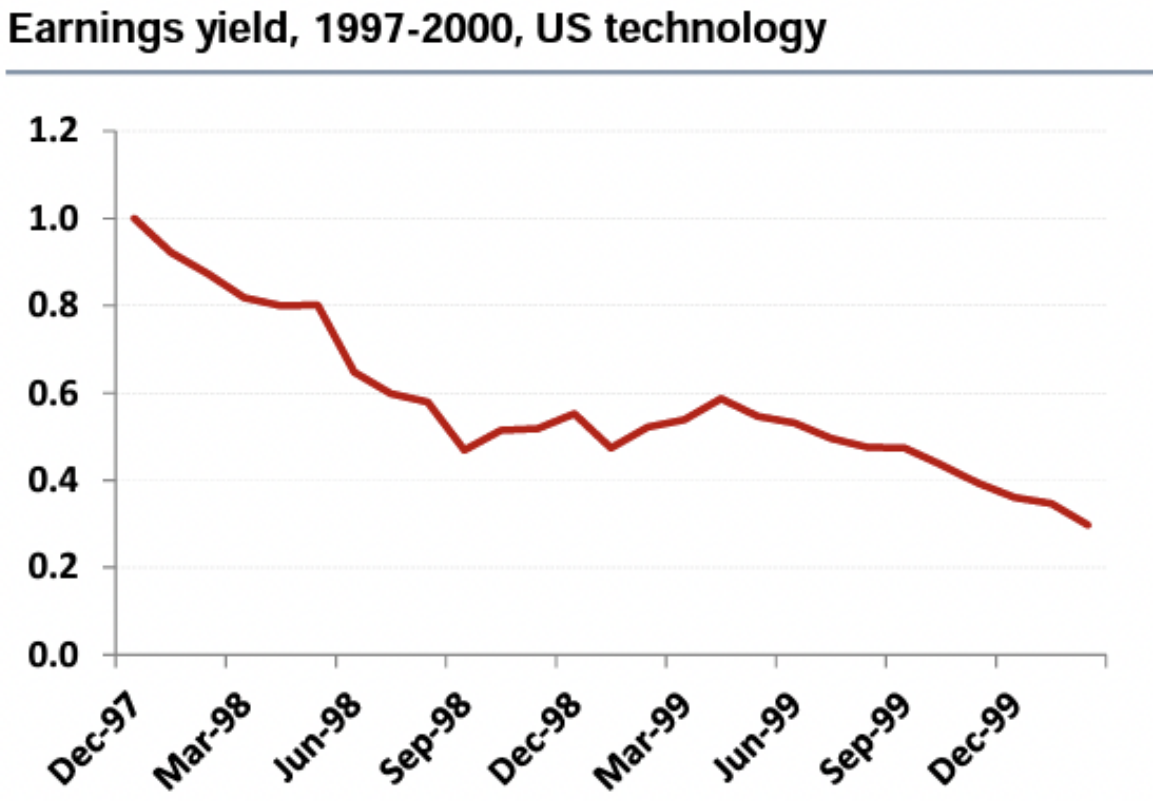

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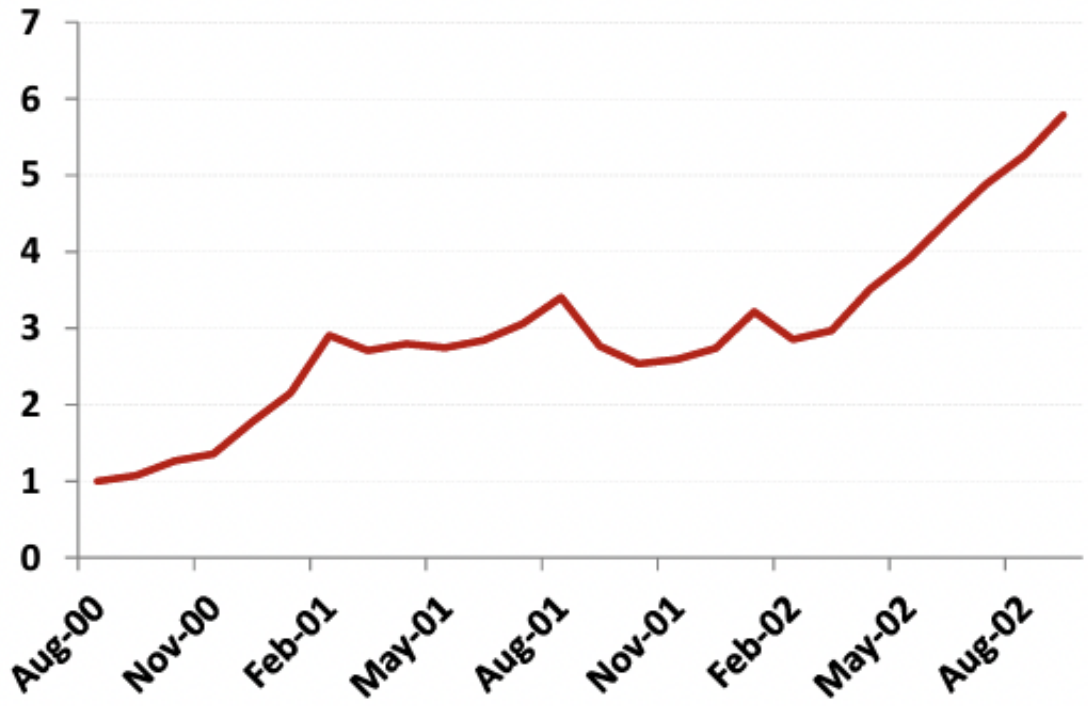
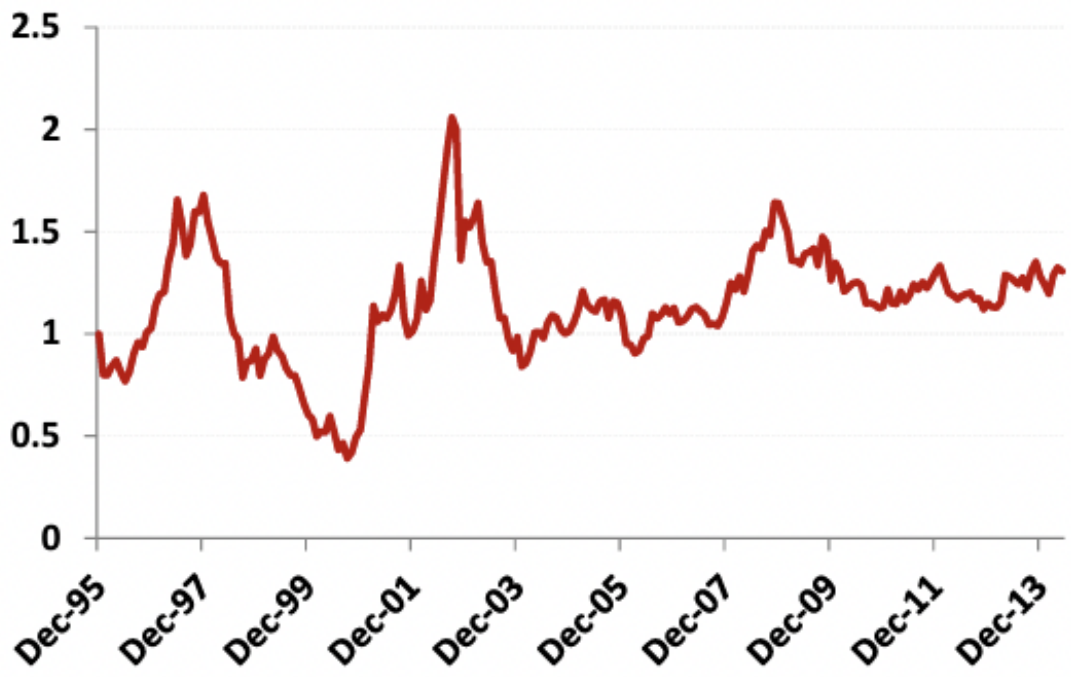
Robustness Check3. How long is long enough?

Earnings yield, 2000-2002, US technology



Earnings yield in US technology sector has never been a good factor



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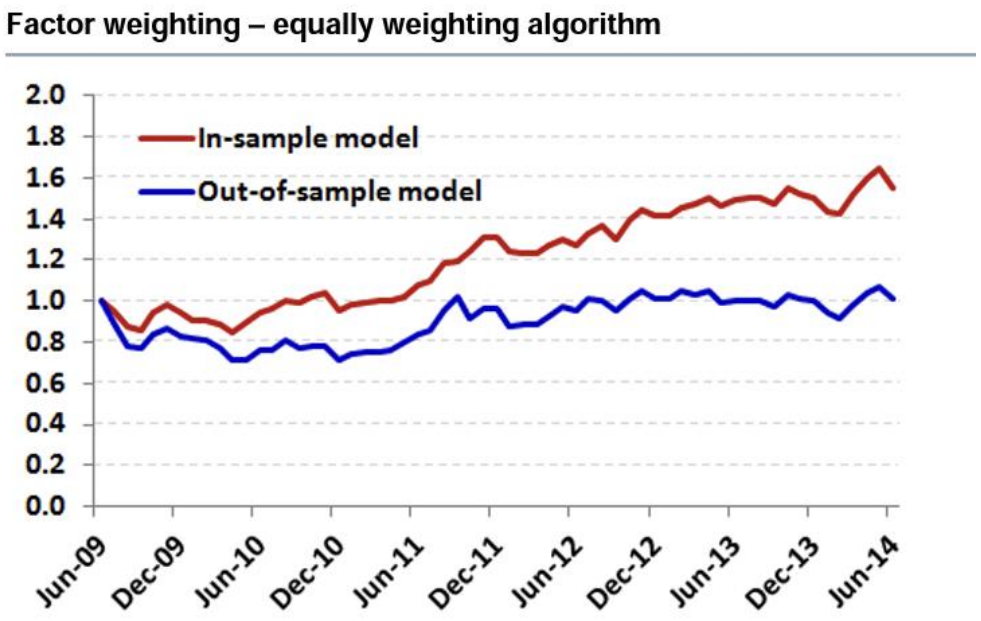
Robustness Check4. Data mining is almost avoidable

- Two factors weighting algorithms
  - **In Sample Model**: select the best factor from each of the six style buckets (value, growth, momentum/reversal, sentiment, quality, and exotic) from 2009-2014 and then backtest the same model over the same period
  - **Out-of-sample model**: from May 31, 2009, at the end of each month, we use rolling 60 months of data to construct our multi-factor model, using data available as of that time



https://newyork.qwafafew.org/wp-content/uploads/sites/4/2015/10/Luo_20150128.pdf

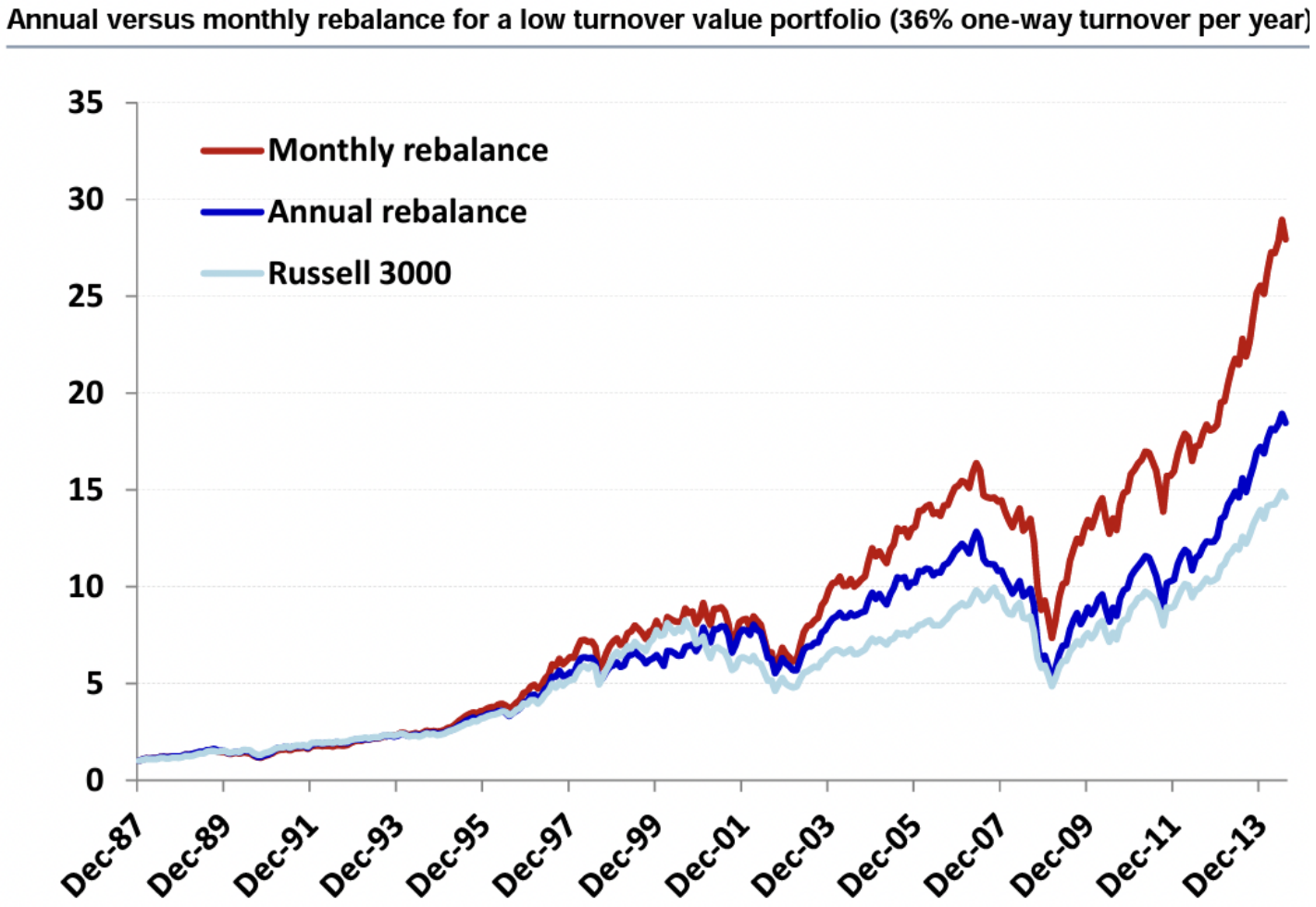
Robustness Check5. The anomaly becomes public after publication

- Many findings have no tomorrow, especially if transaction costs are taken into account (Patton and Weller (2020), A. Y. Chen and Velikov (2020)).

- Agents invest in it, which pushes prices up and the anomaly disappears

- McLean and Pontiff (2016) and Shanaev and Ghimire (2020) document this effect in the US but H. Jacobs and Müller (2020) find that all other countries experience sustained post-publication factor returns

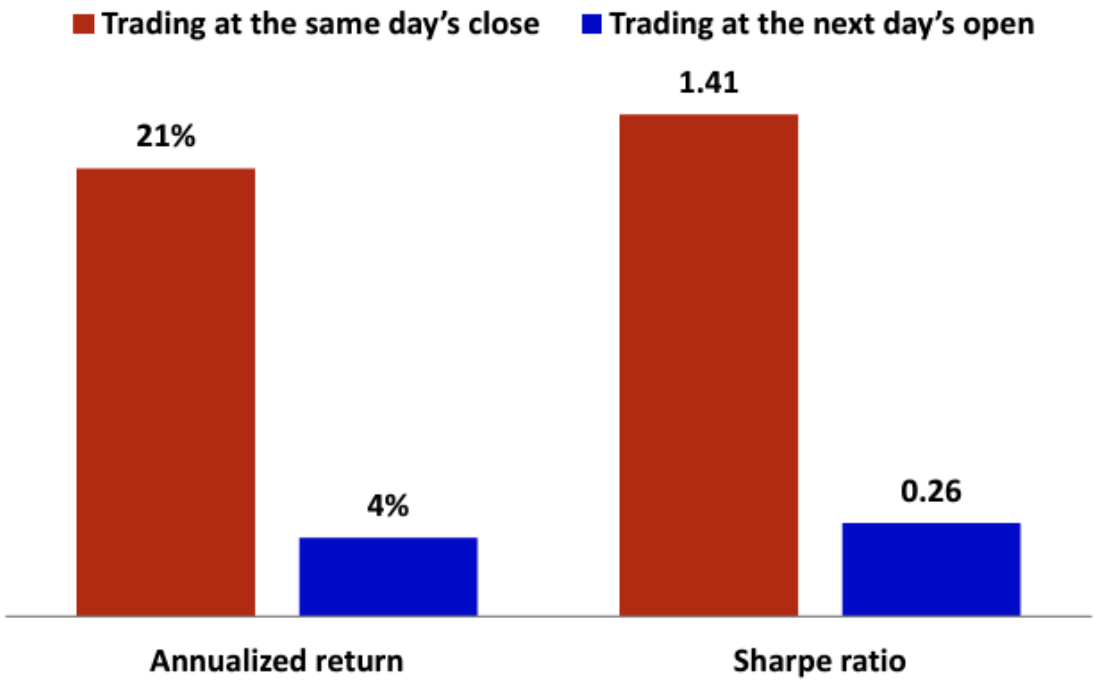
Robustness Check6. Optimal rebalancing period



Robustness Check7. Signal Decay

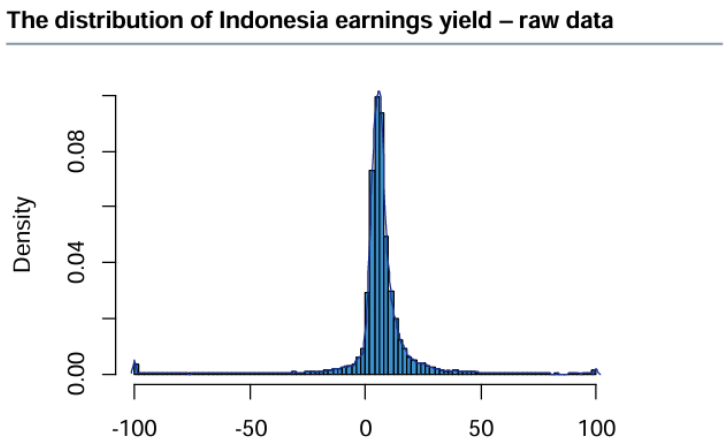
Performance of one day reversal

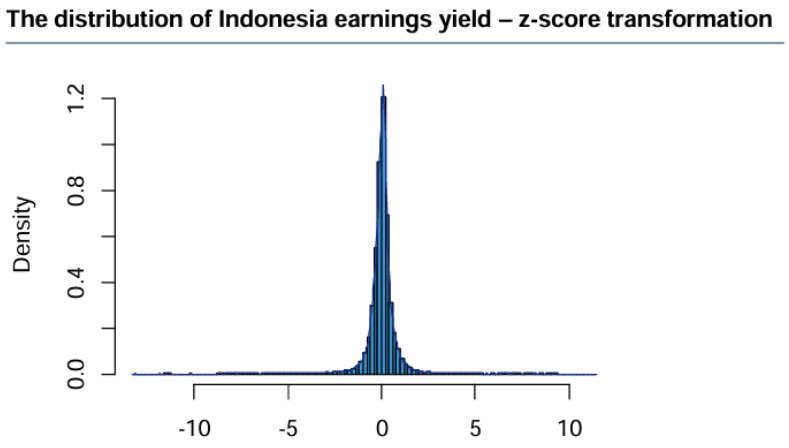
Annualized return and Sharpe ratio

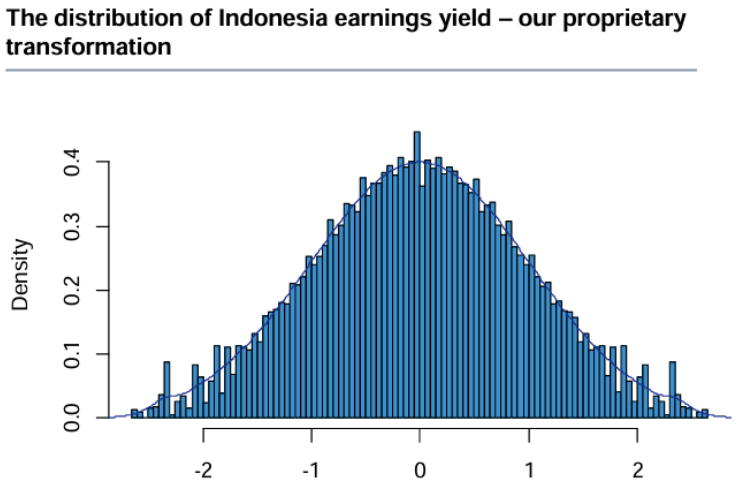


Robustness Check8. Data Transformation

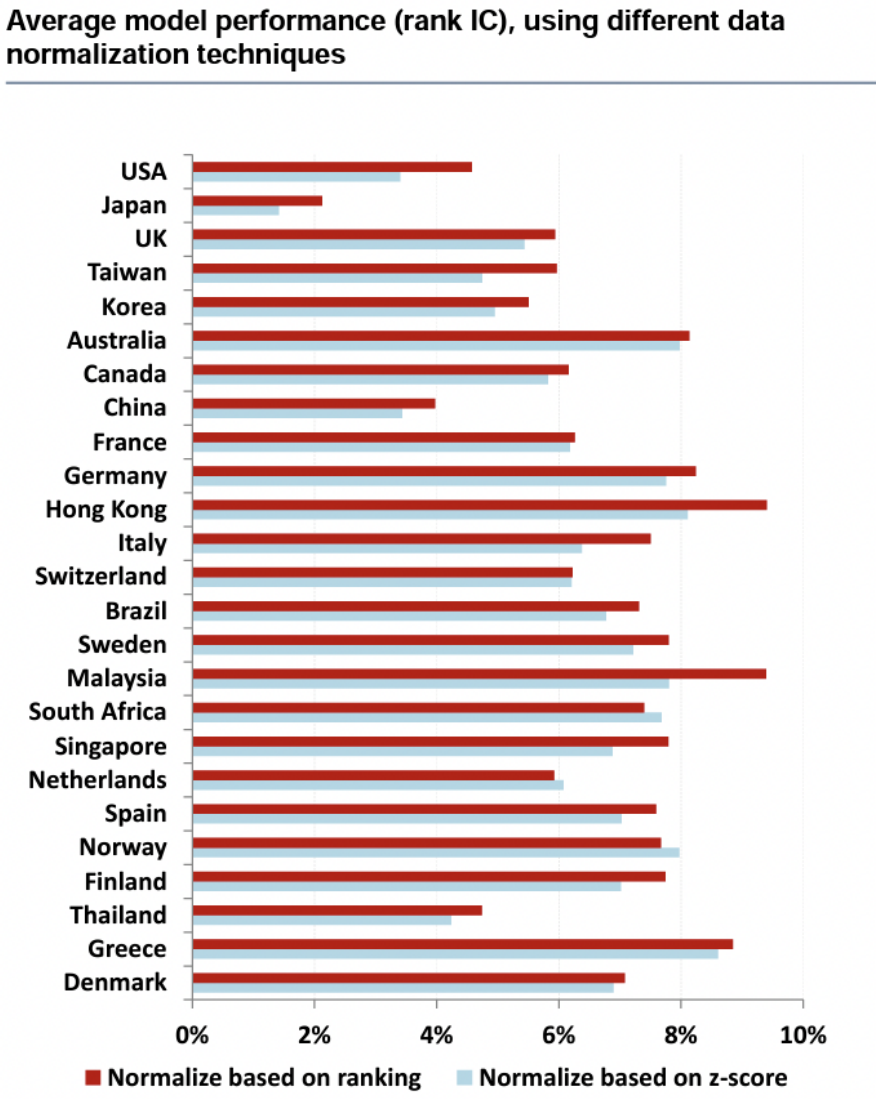
- Four alternative data transformation techniques



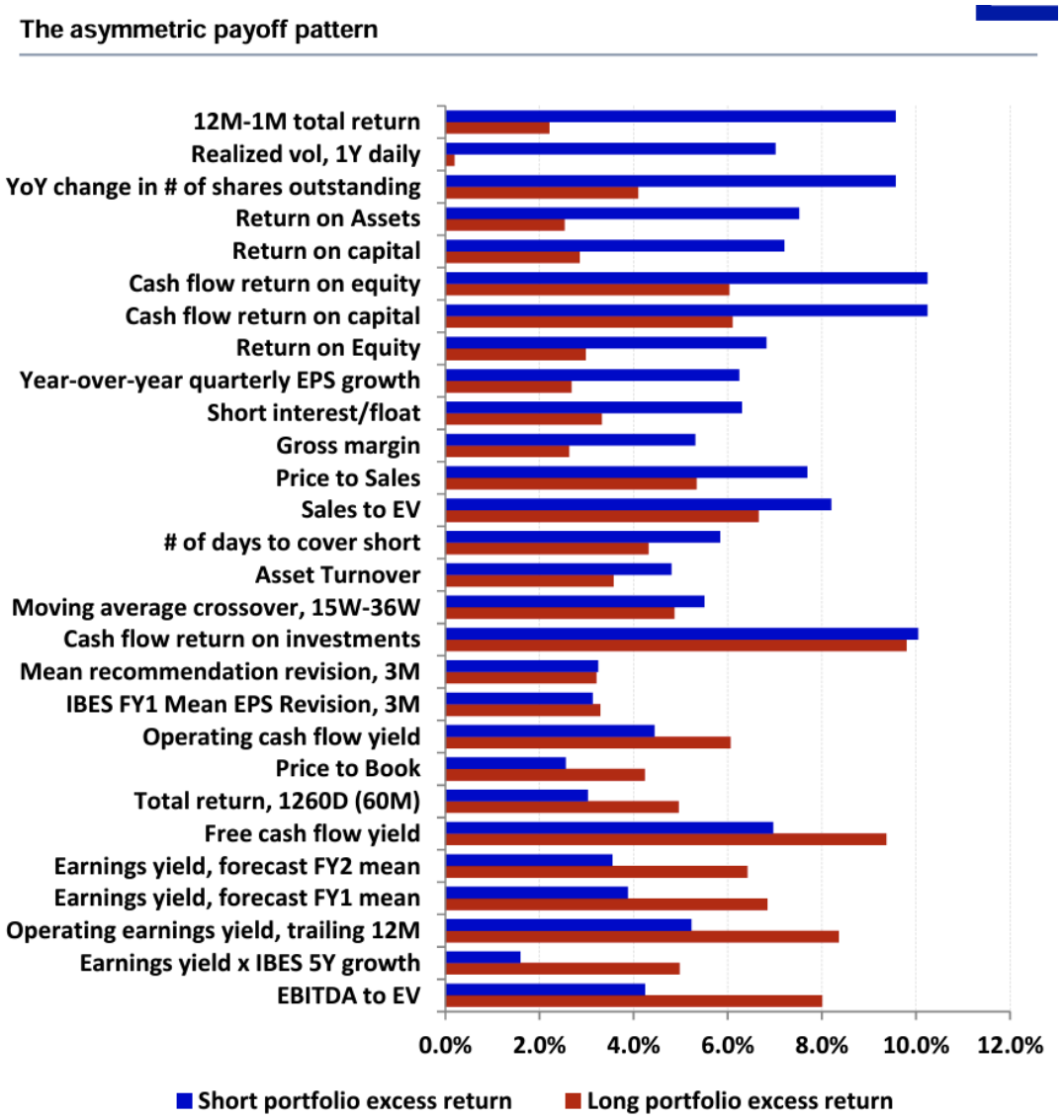


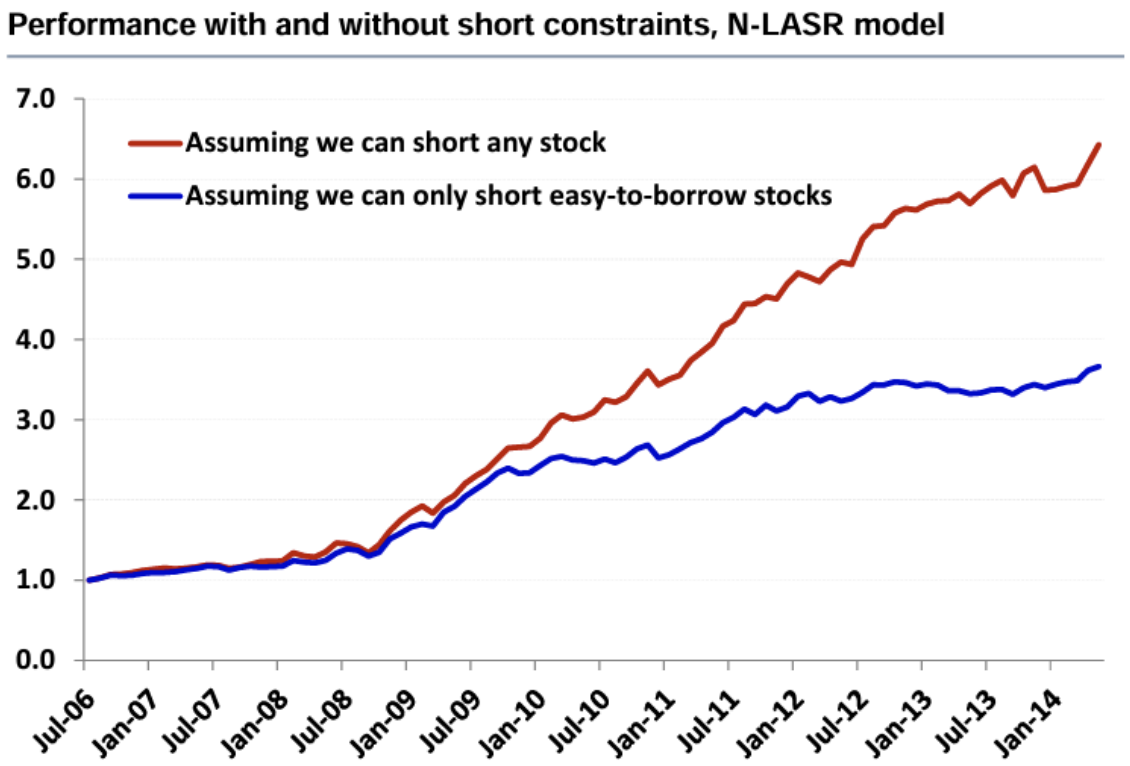


Robustness Check8. Data Transformation



Robustness Check9. Factor Asymmetry and Short Availability





Additional Materials

- Your investment skills are proportional to the reading volume of finance literature & backtesting practices

- But, not all papers are meaningful
  - **Academic Journals:** the Journal of Finance, the Review of Financial Studies, the Journal of Financial Economics
  - **Practitioner Journals:** the Journal of Portfolio Management, the Financial Analysts Journal
  - **Working Papers (SSRN):** Why? Market efficiency, Decaying alphas

Thank You

References

- Kenneth French, https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
- Seven Sins of Backtesting, https://newyork.qwafafew.org/wp-content/uploads/sites/4/2015/10/Luo_20150128.pdf
- ML Factors, http://www.mlfactor.com/
- Equity factor-based investing: A practitioner's guide, https://www.vanguardinvestments.de/documents/institutional/factors-whitepaper-eu.pdf
- Do Portfolio Factors or Characteristics Drive Expected Returns?, https://alphaarchitect.com/2017/10/factors-vs-characteristics/
- Equity Factor Models - Build one in R with a few lines of codes, https://towardsdatascience.com/custom-factor-models-build-your-own-in-r-with-a-few-lines-of-codes-502274ae3624